Rina Bao

Instructor, Harvard Medical School
Scientist, Boston Children's Hospital
Landmark Center, 401 Park Dr., Boston, MA 02115
rina.bao@childrens.harvard.edu | Homepage | Google Scholar

Summary

My research focuses on **AI for Pediatric Healthcare**, integrating multimodal imaging and clinical data to advance diagnosis and prognosis, with an emphasis on **trustworthy and explainable AI for clinical practice**.

Research Contributions

- **Pediatric medical AI**: Developed multimodal learning methods for neonatal brain injury diagnosis, lesion segmentation, prognosis, and neurocognitive outcome prediction.
- **Datasets and benchmarks**: Built and led public HIE neuroimaging benchmarks and datasets, including BONBID-HIE, to support reproducible and clinically meaningful AI evaluation.
- **Trustworthy clinical reasoning**: Advanced explainable and domain-informed AI systems that integrate MRI, clinical variables, and expert knowledge for pediatric healthcare.

Experience & Education

Harvard Medical School, USA & **Boston Children's Hospital, USA**

Instructor 2025 - Present

Postdoctoral Research Fellow 2021 - 2025

Working with Prof. P. Ellen Grant and Prof. Yangming Ou

University of Missouri-Columbia, USA 2021

Ph.D. in Computer Science
Advisor: Prof. Kannappan Palaniappan

Tianjin University, China 2014

B.E. in Communication Engineering

Grants & Fellowships

Charles A. King Trust Research Fellowship **$206,200**

PI: Rina Bao 2025-2027

"Development of neuroimaging biomarkers for 2-year neurocognitive outcomes after neonatal brain injury"

Thrasher Research Fund Early Career Award **$26,750**

PI: Rina Bao 2024-2026

"Development of neuroimaging biomarkers for 2-year neurocognitive outcomes in mild neonatal brain injury"

German Academic Exchange Service (DAAD) AINet Fellowship 2025

Publications

Book Chapter

Bao, R., He, S., Grant, E., Ou, Y., "Linking Deep Learning and MRI to Brain Disorders" in *Handbook of the Biology and Pathology of Mental Disorders.* Springer

Journal

[Scientific Data] Bao, R., Song, Y., Bates, S. V., Weiss, R. J., Foster, A. N., Jaimes, C., Sotardi, S., Zhang, Y., Hirschtick, R. L., Grant, P. E., Ou, Y., "BOston Neonatal Brain Injury Data for Hypoxic Ischemic Encephalopathy (BONBID-HIE): I. MRI and Lesion Labeling", *Scientific Data*, 2025. (IF: 8.7)

[npj Computational Materials] Hajilounezhad, T.*, Bao, R.*, Palaniappan, K., Bunyak, F., Calyam, P., Maschmann, M. R., "Predicting carbon nanotube forest attributes and mechanical properties using simulated images and deep learning", *Nature Partner Journal Computational Materials*, 2021. (* equal contribution) (IF: 13.0)

[IEEE TMI] Bao, R., Foster, A. N., Song, Y., Vyas, R., Kesri, A., Toubal, I. E., Soltani Kazemi, E., Rahmon, G., Kucukpinar, T., Almansour, M., Ho, M.-L., Palaniappan, K., Ninalga, D., Koirala, C. P., Mohapatra, S., Schlaug, G., Wodzinski, M., Muller, H., Ellis, D. G., Aizenberg, M. R., Aydın, M. A., Abdinli, E., Unal, G., Tahmasebi, N., Punithakumar, K., Song, T., Peng, Y., Bates, S. V., Hirschtick, R., Grant, P. E., Ou, Y., "BONBID-HIE 2023: Lesion Segmentation Challenge in BOston Neonatal Brain Injury Data for Hypoxic Ischemic Encephalopathy". *IEEE Transactions on Medical Imaging.* (IF: 9.8)

[MIA] Bao, R., Weiss, R. J., Bates, S. V., Song, Y., He, S., Li, J., Bjornerud, A., Hirschtick, R. L., Grant, P. E., Ou, Y., "PARADISE: Personalized and regional adaptation for HIE disease identification and segmentation", *Medical Image Analysis*, 2025. (IF: 11.8)

[IEEE JSTSP] Bao, R., Zhao, Y., Srivastava, A., Frazier, S., Palaniappan, K., "Lightweight SDE-Net Fusing Model-Based and Learned Features for Computational Histopathology", *IEEE Journal of Selected Topics in Signal Processing*, 2024. (IF: 13.7)

[IEEE TGRS] Bao, R., Palaniappan, K., Zhao, Y., "PointGrad: A Unified Spatially Varying Tensor Gradient Operator for Point Cloud Segmentation", *IEEE Transactions on Geoscience and Remote Sensing.* (IF: 8.6)

[IEEE Sens. J.] Bao, R., Palaniappan, K., Zhao, Y., Seetharaman, G., "GLSNet++: Global and Local-Stream Feature Fusion for LiDAR Point Cloud Semantic Segmentation Using GNN Demixing Block", *IEEE Sensors Journal*, 2024. (IF: 4.7)

[Nature Methods] Maška, M., Ulman, V., Delgado-Rodriguez, P., Gómez-de-Mariscal, E., Nečasová, T., Guerrero Peña, F. A., Ing Ren, T., Meyerowitz, E. M., Scherr, T., Löffler, K., Mikut, R., Guo, T., Wang, Y., Allebach, J. P., Bao, R., Al-Shakarji, N. M., Rahmon, G., Toubal, I. E., Palaniappan, K., Lux, F., Matula, P., Sugawara, K., Magnusson, K. E. G., Aho, L., Cohen, A. R., Arbelle, A., Ben-Haim, T., Riklin Raviv, T., Isensee, F., Jäger, P. F., Maier-Hein, K. H., Zhu, Y., Ederra, C., Urbiola, A., Meijering, E., Cunha, A., Muñoz-Barrutia, A., Kozubek, M., Ortiz-de-Solórzano, C.,"The Cell Tracking Challenge: 10 years of objective benchmarking", *Nature Methods*, 2023. (IF: 51.7)

[JAMIA Open] Hsiao, C.-H., Foster, A. N., McDonald, S. A., Vyas, R., Ashraf, A., Bao, R., Tran, L., Kesri, A., Darzidehkalani, E., Soldatelli, M. D., Auman, J. O., Soul, J. S., Chalak, L. F., Cotten, C. M., Shankaran, S., Laptook, A. R., Grant, P. E., Ou, Y., "Data Harmonization Framework for Neonatal Hypoxic-Ischemic Encephalopathy Studies", *Journal of the American Medical Informatics Association Open*, 2025 (IF: 3.2)

Conference

[ICML'25] Bao, R., Dong, S., Chen, Z., He, S., Grant, E., Ou, Y., "Visual and Domain Knowledge for Professional-level Graph-of-Thought Medical Reasoning", *International Conference on Machine Learning*, 2025. (**Spotlight**, **Top 2.6%**)

[MICCAI'25] Rao, A., Shukla, A., Jia, B., Ou, Y., Bao, R., "Spatial Prior-Guided Boundary and Region-Aware 2D Lesion Segmentation in Neonatal Hypoxic Ischemic Encephalopathy", *MICCAI*, 2025.

[AIPR'25] Hatuwal, B. K., Bao, R., Ho, M.-L., Palaniappan, K., "HE3D-Net: Hypoxic Ischemic Encephalopathy Diagnosis and Lesion Segmentation Using 3D Heterogeneous Ensemble Models", *Applied Imagery Pattern Recognition Workshop*, Springer, 2025.

[ICCV W'21] Bao, R., Al-Shakarji, N. M., Bunyak, F., Palaniappan, K., "DMNet: Dual-stream marker guided deep network for dense cell segmentation and lineage tracking", *Int. Conf. on Computer Vision (ICCV) Computer Vision for Automated Medical Diagnosis (CVAMD) Workshops*, 2021. (**Best Paper Award**)

[ISBI'26] Rao, A., Shukla, A., Jia, B., Siddharth, S., Bao, R., "Obscuration to Clarity: Bone Suppression for Enhanced Localization in Pneumothorax Segmentation of Chest Radiographs", *ISBI, 2026.* (**Oral**)

[ISBI'25] Bao, R., He, S., Grant, E., Ou, Y., "AGE2HIE: Transfer Learning from Brain Age to Predicting Neurocognitive Outcome for Infant Brain Injury", *ISBI, 2025.*

[ISBI'25] He, S., Bao, R., Grant, P. E., Ou, Y., "Relation U-Net", *ISBI*, 2025

[ISBI'25] Li, J., He, S., Bao, R., Bjørnerud, A., Grant, E., Ou, Y., "Using Off-Target-Disease Data to Improve Target-Disease Lesion Segmentation", *ISBI*, 2025.

[IEEE AIPR'19] Bao, R., Palaniappan, K., Zhao, Y., Seetharaman, G., Zeng, W., "GLSNet: Global and local streams network for 3D point cloud classification", *IEEE Applied Imagery Pattern Recognition Workshop*, 2019. (**Oral**)

Conference Abstracts

[PAS'26] Bao, R., Hsiao, C.-H., Bates, S. V., Weiss, R., Gollub, R., Grant, E., Ou, Y., "Anatomy-Normalized ZADC Maps Improves Detection of Neonatal Brain Injury in HIE", Pediatric Academic Societies Meeting (**PAS**), 2026 (Oral).

[PAS'26] Bao, R., Grant, E., Ou, Y., "Agentic Artificial Intelligence for Hypoxic-Ischemic Encephalopathy Prognosis through Clinically Guided Reasoning on Neonatal MRI", Pediatric Academic Societies Meeting (**PAS**), 2026 (Oral).

[SPR'23] Bao, R., Grant, E., Ou, Y., "2-Year neurocognitive outcome prediction of Hypoxic Ischemic Encephalopathy in neonates with brain MRIs", The Society for Pediatric Radiology (**SPR**) Annual Meeting, 2023.

[Nutrition'23] Bao, R., Wu, D., Vyas, R., Kuchan, M., Lasekan, J., Sutton, B., Grant, E., Morton, S., Ou, Y. "Maternal Dietary Nutrition Interact With Demographic and Socioeconomic Information To Impact Offspring Neurocognition by Early School Age", **Nutrition**, 2023.

[PAS'23] Bao, R., Grant, E., Ou, Y., "Clinical Data Predicting Neonatal and 2-year Outcome Following Hypoxic Ischemic Encephalopathy", Pediatric Academic Societies Meeting (**PAS**), 2023.

[PAS'25] Ou, Y., Kesri, A., Bao, R., Hsiao, C.-H., Vyas, R., Hussain, M. A., McDonald, S., Auman, J., Foster, A., Darzi, E., Soldatelli, M., Shankaran, S., Laptook, A., Cotten, M., Grant, E., "Combining 47 clinical and neuroimaging variables to predict adverse 18-22-month outcomes for Hypoxic Ischemic Encephalopathy in Neonatal Research Network trials", Pediatric Academic Societies Meeting (**PAS**), 2025.

[ASPNR' 25] Kesri, A., Bao, R., Hsiao, C.-H., Vyas, R., Hussain, M. A., McDonald, S., Auman, J., Shankaran, S., Laptook, A., Cotten, M., Grant, E., Ou, Y., "The role of expert MRI scores in predicting adverse 18-22-month outcomes for Hypoxic Ischemic Encephalopathy in Neonatal Research Network trials", Annual Meeting of American Society of Pediatric Neuroradiology (**ASPNR**), 2025.

Under Review

[Science Advances] Bao, R., Zhao, Y., Ou, Y., Palaniappan, K., "Brain-Inspired Visual Primitives Enable Cross-Domain Learning from Few Examples", major revision at *Science Advances.*

[npj Science of Food] Bao, R., Wu, D., Vyas, R., Kuchan, M. J., Lasekan, J. B., Leyshon, B., Gopinath, A., Sutton, B. P., Grant, P. E., Morton, S. U., Ou, Y., "Maternal nutrition and sociodemographic factors determinants jointly impact neurocognition of breastfed offspring at early school age: A pilot machine learning study", under review at *npj Science of Food.*

[Epilepsia] Kesri, A.*, Bao, R.*, Valery, C., Iannotti, I., McLaren, J., Pinto, A. L., Comi, A. M., Ou, Y., "Machine Learning Could Pre-Symptomatically Predict Seizure Onset by 2 Years for Sturge-Weber Syndrome", under review at *Epilepsia*. (* equal contribution)

[AI in Radiology] Bao, R., Foster, A. N., Song, Y., Vyas, R., Xu, R., Piao, Z., Choi, J., Gu, Y., Ninalga, D., Choi, J. W., Hatuwal, B. K., Kazemi, E. S., Palaniappan, K., Rao, A., Shukla, A., Bhargava, J., Dhouib, H., Pan, Z., Liang, X., Yıldırım, A. S., Şahin, Y. H., Baek, S., Morton, S. U., Bates, S. V., Hirschtick, R., Grant, P. E., Ou, Y. "BONBID-HIE 2024: Lesion Segmentation and Outcome Prediction from Boston Neonatal Brain Injury MRI in Hypoxic-Ischemic Encephalopathy — Review and Results", under review at *AI in Radiology*.

[MIA] Darzi, E., Hsiao, C.-H., Bao, R., Weiss, R. J., Bates, S. V., Song, Y., He, S., Li, J., Bjornerud, A., Hirschtick, R. L., Grant, P. E., and Ou, Y., "SparseMed3D: Foundation Models for Sparse Instance Medical Segmentation", under review at Medical Image Analysis.

Preprints

Bao, R., Ou, Y., "BOston Neonatal Brain Injury Dataset for Hypoxic Ischemic Encephalopathy (BONBID-HIE): Part II. 2-year Neurocognitive Outcome and NICU Outcome", arXiv:2411.03456.

Bao, R., Darzi, E., He, S., Hsiao, C.-H., Hussain, M. A., Li, J., Bjornerud, A., Grant, E., and Ou, Y.,"Foundation AI Model for Medical Image Segmentation", arXiv:2411.02745.

He, S., Bao, R., Grant, P. E., and Ou, Y., "U-Netmer: U-Net meets Transformer for medical image segmentation", arXiv:2304.01401.

He, S., Bao, R., Li, J., Stout, J., Bjornerud, A., Grant, P. E., and Ou, Y., "Computer-vision benchmark segment-anything model (SAM) in medical images: Accuracy in 12 datasets", arXiv:2304.09324.

Datasets, Benchmarks & Challenges

BONBID-HIE Dataset and Benchmarks

Built and organized public neonatal HIE MRI benchmarks and challenges used by over 300 researchers from over 20 countries, with datasets downloaded over 4,000 times.

- 1^{st} BONBID-HIE Challenge for HIE Lesion Segmentation in MICCAI 2023.
- 2^{nd} BONBID-HIE Challenge for HIE Lesion Segmentation and Outcome Prediction in MICCAI 2024.

Professional Services

Organizer of Pediatric Brain Analysis Workshop in ISBI 2026.

Editor for **MICCAI Challenge LNCS proceedings**

- "AI for Brain Lesion Detection and Trauma Video Action Recognition", 2025.

Reviewer for npj Digital Medicine, Frontiers in Neuroscience, Scientific Reports, Pattern Recognition, IEEE TIP, IEEE TBE, IEEE TNNLS, CVPR, MICCAI, ISBI, etc.

Internship

Microsoft Research Asia — Intelligent Multimedia Group — 2018

Developed a real-time video pose estimation and tracking system, **showcased at Microsoft TechFest 2018 (Redmond) and integrated into a Microsoft product**.

Teaching Experience

Teaching Assistant, Department of Electrical Engineering and Computer Science, University of Missouri-Columbia

- **CMP_SC 3050: Advanced Algorithm Design** — Fall 2016
  (>80 undergraduate students)
- **CMP_SC 4850/7850: Computer Networks I** — Spring 2016
  (> 50 undergraduate & graduate students)

Student Advising

Shilong Dong, New York University, USA. Now Machine Learning Engineer at NVIDIA.
Project: “Clinical Reasoning on Neonatal Brain MRIs”. Outcome: ICML 2025 Spotlight.

Dean Ninalga, University of Toronto, Canada
Project: “Lesion Detection with Clinical Priors”.

Di Wu, Johns Hopkins University, USA. Now Deep Learning Architect at AWS.
Project: “Feature Selection in Clinical Data”. Outcome: manuscript under review at npj Science of Food.

Shuyi Fan, University of Massachusetts Lowell, USA.
Project: “Agentic AI for CT Understanding”.

Ankush Kesri, Northeastern University, USA. Now a Data Science Manager at BCH.
Project: “Small and Diffuse Lesion Segmentation”

Dat Duong, University of Massachusetts Amherst, USA.
Project: “Loss Functions in Small and Diffuse Lesions”

Ananya Shukla, Plaksha University, India. Outcome: MICCAI 2025 and ISBI 2026 Oral.
Project: “Spatial Prior-Guided Boundary and Region-Aware 2D Lesion Segmentation”.
Project: “Bone Suppression for Pneumothorax Segmentation of Chest Radiographs.”
Amog Rao, Plaksha University, India. Outcome: MICCAI 2025.
Project: “Spatial Prior-Guided Boundary and Region-Aware 2D Lesion Segmentation”.

Awards & Honors

Trainee Travel Award, PAS — 2026

Nomination for Rising Star in Healthcare, Boston Children’s Hospital — 2025

Best Paper Award
ICCV Computer Vision for Automated Medical Diagnosis (CVAMD) Workshop — 2021

Second Place
ISBI 2021 Cell Segmentation and Tracking Challenge — 2021

Top Graduate Team Prize
NeurIPS SpaceNet7 Challenge (Team Leader, Among over 300 entries) — 2020

First Prize (Top 1.06%)
China Undergraduate Mathematical Contest in Modeling — 2012